

**INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS
ACADEMY**

IPv4 Addressing and Subnet Design

Laboratory 1

Student Practical Laboratory Manual

Course	CORE102 TCP IP Networking and Linux Network Tools
Instructor	Aminu Idris AMCPN
Maximum score	100 marks
Gradebook category	Practical Laboratories
Estimated duration	4 to 5 hours
Platform	Kali Linux or Ubuntu Linux in an authorised laboratory environment
Version	September 2026

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Submission date 09/30/2026

I started with a baseline capture of the host's network state. The active interface was eth0 at 192.168.56.102/24. No default gateway existed on this host-only VM, so ip route get 8.8.8.8 returned "Network is unreachable" rather than selecting a source address.

I converted 192, 168, 10 and 25 to binary using Python's :08b, confirming 192.168.10.25 is 11000000.10101000.00001010.00011001. I then recorded each test address's historical class and default mask, and re-analysed it under the given CIDR prefix. Class only describes the mask the address would have had under classful rules — it does not constrain the prefix. For example, 10.24.6.9 is class A by first octet (default /8) but deployed as /20, giving network ID 10.24.0.0 and broadcast 10.24.15.255.

I used Python's ipaddress module to subdivide 192.168.10.0/24 into /25, /26 and /27 subnets and cross-checked every value against manual work. Block sizes are 128, 64 and 32. The address 192.168.10.173 falls in the sixth /27 subnet — 192.168.10.160/27 — because .160 is the largest block start below .173 and .191 is its broadcast.

For Part 4 I created a temporary dummy interface core102lab0 and assigned 192.168.10.10/25 and 192.168.10.130/25. Both addresses and their connected routes were verified, and pings succeeded locally because the kernel answers its own interface addresses. The interface was then deleted and cleanup confirmed.

For Part 5 I designed a VLSM plan on 192.168.50.0/24: Training hall /25, Staff /26, Cyber range /27, Management /28, Router link /30, leaving 192.168.50.244–.255 unallocated.

Finally, subnet_report.py accepted valid inputs with exit status 0 and rejected a missing prefix with exit 1 and a malformed address with exit 2.